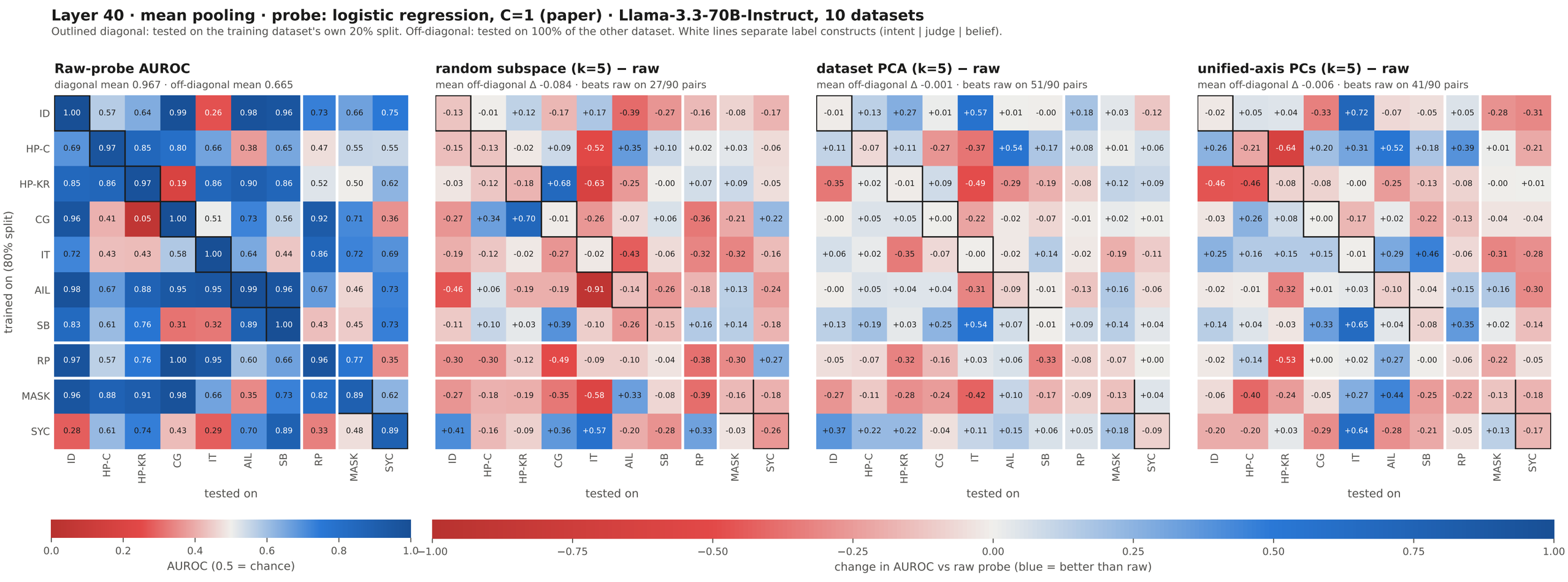

Layer 40 · mean pooling · probe: logistic regression, C=1 (paper) · Llama-3.3-70B-Instruct, 10 datasets
Outlined diagonal: tested on the training dataset's own 20% split. Off-diagonal: tested on 100% of the other dataset. White lines separate label constructs (intent | judge | belief).
Raw-probe AUROC
diagonal mean 0.967 · off-diagonal mean 0.665
random subspace (k=5) − raw
mean off-diagonal Δ -0.084 · beats raw on 27/90 pairs
dataset PCA (k=5) − raw
mean off-diagonal Δ -0.001 · beats raw on 51/90 pairs
unified-axis PCs (k=5) − raw
mean off-diagonal Δ -0.006 · beats raw on 41/90 pairs
trained on (80% split)
tested on
ID
HP-C
HP-KR
CG
IT
AIL
SB
RP
MASK
SYC
AUROC (0.5 = chance)
change in AUROC vs raw probe (blue = better than raw)